

Backup Infrastructure — Recovery, Cleanup & Redesign

Full report — incident through target-state · for review with T-Net

Date: 2026-07-23 Prepared by: Ian Lalonde, KOR Period: 2026-07-07 → 07-23

Executive summary

An expired vCenter certificate silently stopped all backups for **~8 days** (2026-07-07 → 07-15); the condition went undetected by monitoring and was found only during a manual console review. Recovery was complicated by a same-window deduplication rollout on the file server and a job design (reverse-incremental on an under-sized repository) that could not complete. Over 2026-07-15 → 07-23, KOR recovered all services, upgraded the platform, reclaimed ~20 TB, restore-tested every copy, and re-architected the environment to a self-verifying, ransomware-hardened design using the existing Enterprise Plus licence.

- Service:** All backups restored and healthy; off-site copy of the file server seeding now
- Verification:** First-ever restore tests performed — 2 of 3 pass; 3rd resolved via platform upgrade
- Space reclaimed:** ~20 TB on production; ~36 TB more identified/pending (E: chain + Synology archive)
- Platform:** Veeam upgraded 12.0.0.1420 → 12.3.2; deduplicated file-level restore now working

1. Incident & root cause

Date	Event
Jul 7 ~17:00	Last successful backups (confirmed by restore-point files).
Jul 7/8	vCenter TLS certificate expires; Veeam pins the thumbprint → all jobs + discovery fail. Backup coverage stops. No alert reaches KOR.
Jul 13	Windows deduplication enabled on KOR-FS01 (E:) — rewrites ~10 TB, purges existing VSS shadow copies (uncommunicated).
Jul 15	KOR discovers the outage via manual console review; re-accepts the new certificate; restarts jobs. 8-day gap ends.
Jul 15–19	Restarted reverse-incremental job cannot fit the dedup churn in the 25 TB repository; stopped per Veeam KB1727.
Jul 19–23	File server re-protected (forward-incremental → Synology); platform upgraded; ~20 TB reclaimed; restore-tested; redesign built.

Two independent root causes: (1) an expired certificate plus monitoring that did not detect an 8-day total outage (provider-side); (2) a deduplication rollout executed without assessing backup impact — shadow-copy loss and a disk-full cascade (change-control side). The July 8–13 window is exposed by their intersection (backups dead, then shadow copies purged).

2. Recovery — completed

#	Action	Result
1	Re-accepted the new vCenter certificate; restarted the job chain	Jobs reconnected
2	Stopped the non-viable reverse-incremental run (KB1727); confirmed FS01 snapshot removed (dual-verified)	Clean, no orphan snapshot

3	Reclaimed system-drive space on BK01; Windows updates + reboot; verified services	Healthy
4	New dedicated KOR-FS01 job — forward incremental → Synology	5.79 TB, 126 MB/s, ~14 h, 0 errors
5	Upgraded Veeam 12.0.0.1420 → 12.3.2 (config backup + VM snapshot first)	Clean; 3 yrs of patches
6	Added KOR-APP01 (Server 2025) as mount server for the Synology repo	Deduplicated file-level restore now works
7	Configured Veeam email notifications to KOR	Verified delivering (07-21)
8	Re-enabled small-VM backups + off-site copy; built new FS01 off-site copy job	All tiers live; FS01 off-site seeding

Restore testing — first ever performed (2026-07-20)

Scenario	Result	Finding
FS01 file-level from local pre-dedup chain (E:)	PASS	Local history genuinely restorable
FS01 file-level from off-site Cloud Connect	PASS	Off-site copy restorable over the WAN
FS01 file-level from new (deduplicated) Synology backup	FAIL → FIXED	Needed a Server 2025 mount server; resolved by adding APP01 (item 6). Re-test PASS.

3. Cleanup & space reclamation — audit trail

Item removed	Size	Location	Status
BMZ-HO-FS007 (retired file server)	~10.5 TB	UC3200 (prod)	Deleted
BMZ-HO-DC005 (retired)	~9 TB	UC3200 (prod)	Deleted
BMZ-HO-APP002 (retired)	~0.6 TB	UC3200 (prod)	Deleted
BMZ-HO-RDS001 (retired)	~0.1 TB	UC3200 (prod)	Deleted
Orphaned Dec-2025 <code>.vrb</code> files	~46 GB	E: repo	Deleted
Orphaned <code>Kor-VMs</code> backup (corrupted FS007 stub)	~48 KB	E: (orphaned)	Deleted
BMZ-HO-APP003 / APP004 backups	small	E: + Cloud	Deleted
Stale duplicate Kor-RDS01 (Jan/Feb identity split)	small	E: + Cloud	Deleted
Old FS01 reverse-incremental chain	~23 TB	E: repo	Pending — after off-site verified

Old FS01 copy (pre-dedup, Jul 7)	several TB	Cloud Connect	Pending — after off-site verified
ManualNew archive → relocate to E:	~12.7 TB	Synology → E:	Pending — retained as archive #1

Result: ~20 TB reclaimed on production (UC3200 7.5 → 27 TB free, i.e. 15% → 47% free). A further ~36 TB is identified and staged for reclaim once the off-site copy is verified (the safety net is retained until the replacement is proven). Retired VMs, orphaned records, and duplicate identities are removed; ManualNew (a March-2026 point-in-time) is deliberately kept as the first archive.

4. Redesign — target architecture

Production (vSphere / UC3200) → **Veeam BK01 (12.3.2)** →

- ① **Primary** → **Synology (F:)** forward-incr + weekly synthetic full · mount server APP01 · Health Check + Threat Hunter on
- ② **Off-site** → **Cloud Connect (T-Net)** Backup Copy over the wire (WAN accel to be enabled)
- ③ **Archive** → **E: (repurposed)** point-in-time fulls via GFS + Export · immutable target (phase 2)
- ④ **Verification** → **SureBackup virtual lab** boots & tests backups on schedule

Design principle: **forward incremental + synthetic fulls** everywhere (measured ~4× faster, ~1/3 the space of reverse-incremental on this data), and a genuine **3-2-1** — primary (Synology), off-site (Cloud), point-in-time archive (E: → immutable).

5. Feature roadmap — Enterprise Plus, fully leveraged

Capability	Veeam feature	Status
Deduplicated file-level restore	APP01 (Server 2025) mount server	DONE
Off-site copy of the file server	Backup Copy → Cloud Connect	SEEDING
Failure alerting	Email notifications	DONE
Automated recovery verification	SureBackup (isolated virtual lab boot/test)	TO IMPLEMENT
Backup integrity assurance	Health Checks (periodic auto re-verify)	TO IMPLEMENT
Ransomware — malware scanning	Veeam Threat Hunter (installed w/ 12.3)	TO ENABLE
Ransomware — anomaly detection	Suspicious-activity alerts (12.3)	TO ENABLE
Ransomware — immutable archives	Object-lock storage / hardened repo	PHASE 2
Point-in-time / clean-room history	GFS retention + Export Backup	TO IMPLEMENT
Fast recovery (RTO)	Instant VM Recovery (validate; APP01 cache)	TO VALIDATE
WAN throughput to off-site	WAN Acceleration (T-Net cloud-side)	T-NET
Storage efficiency	Fwd-incr + synthetic full + ReFS/XFS fast-clone + BitLooker	PARTIAL

Regular restore drills	Quarterly file-level, annual full-VM / IVMR	TO SCHEDULE
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6. Current protection status

System	Local backup	Off-site	Point-in-time
KOR-FS01 (file server)	Synology (current) + E: chain (safety net)	Seeding (old Jul-7 copy remains meanwhile)	ManualNew Mar-2026
KOR-APP01, DC01, RDS01	E: (current, nightly)	Cloud Connect (current)	ManualNew Mar-2026
vCenter	E: (current)	Cloud Connect (current)	—

7. Outstanding & T-Net items

KOR to complete (self-driven):

- Finish FS01 off-site seed → verify restore → retire the old E: chain (~23 TB) and old cloud FS01 copy.
- Relocate ManualNew archive to E; stand up SureBackup, Health Checks, Threat Hunter, anomaly alerts, GFS archives.
- Convert small-VM jobs to forward incremental; validate Instant VM Recovery; retire remaining BMZ-HO-APP003/004; resolve RDS01 duplicate identity.

T-Net (retained dependencies / accountability):

- **Monitoring/alerting SLA** — why an 8-day total outage produced no alert; how job failures will reach KOR going forward.
- **vCenter certificate ownership** — logged renewal reminder ahead of 2028-07-14.
- **Patch policy** for backup infrastructure (platform was 3 years behind).
- **WAN Acceleration** on Cloud Connect (target-side accelerator) + confirm any tenant bandwidth cap (seed observed at ~8 MB/s on a 182 Mbps link) + storage quota headroom.
- **Licence** — confirm Enterprise Plus rental covers continued use through these changes.
- Confirm origin/purpose of the ManualNew job.

KOR Structural Operations, 2026-07-23. Compiled from live console, repository, filesystem, and Veeam-database evidence, 2026-07-15 → 07-23. Space figures verified in vSphere and the Veeam console. Deletions listed in §3 were performed after data-ownership confirmation and, where applicable, dual verification (e.g., the FS007 shared-disk check). Underlying logs, restore-point listings, and screenshots available on request.